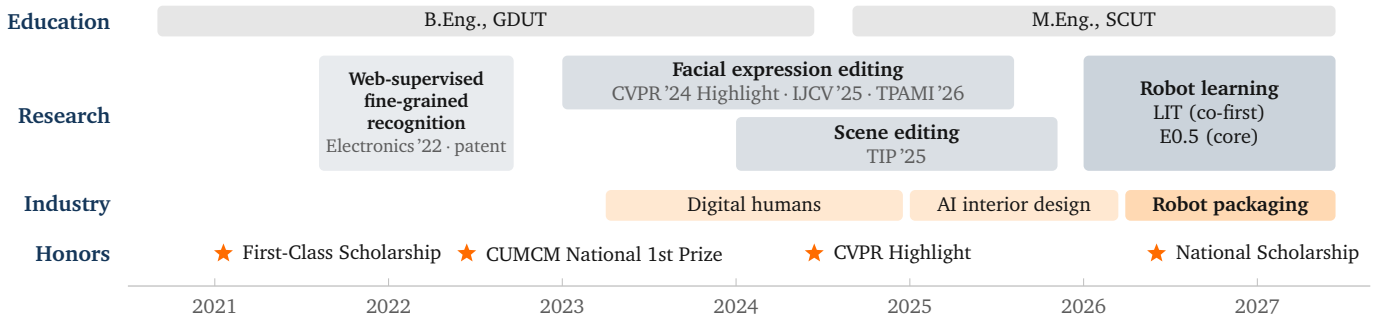
Jianman Lin

Embodied Intelligence · Multimodal Learning · Generative Modeling

linjianmancjx@gmail.com | Homepage | Google Scholar

Research Profile

In my second undergraduate year, I joined **Prof. Tianshui Chen**'s group and started research with **web-supervised fine-grained image recognition**. I then turned to **generative models**, working on **facial expression editing** and **scene editing**, alongside industry work on digital humans and AI interior design. On real robots, I found that VLA policies rely on irrelevant visual cues; this led to **LIT**, and I also contributed to the 3D world–action model **E0.5**. Publications include first/co-first-author papers in **TPAMI, IJCV and TIP** and a **CVPR Highlight**.



Education

South China University of Technology (SCUT) — Guangzhou, China
M.Eng. in Electronic Information — Sep. 2024 – June 2027 (expected)
Advisors: Prof. Tianshui Chen and Prof. Chunmei Qing.

Guangdong University of Technology (GDUT) — Guangzhou, China
B.Eng. in Industrial Engineering (GPA 87.2/100) — Sep. 2020 – June 2024

Research Experience

LIT: Generalizable Robot Foundation Models [1] — 2026 – Present

- **Motivation.** Robots that succeed in their training scene often fail under visual distribution shifts (e.g., new lighting or camera views): they learn visual shortcuts rather than where the task is. *How can action learning and visual conditioning be structured to mitigate these shortcuts while preserving the spatial information needed to act?*
- **Method.** Two stages: an image-free action prior conditioned on the target end-effector pose, then a pose-supervised latent interface as the only path from vision to action.
- **Results.** No loss on LIBERO; zero-shot LIBERO-Plus success **+3.9–10.7 points** across four VLA and world-action architectures; **+13.3–16.7 points** under unseen visual conditions on three real-robot tasks.
- **My role.** Identified the problem, proposed and implemented the method, ran all simulation experiments and wrote the paper; the co-first author led real-robot experiments; Prof. Jiafei Duan advised.

Speech-Preserving Facial Expression Manipulation [3, 5, 6] — 2023 – 2025

- **Motivation.** Speech-preserving facial expression manipulation (SPFEM) changes a talking face's emotion while keeping its lips in sync with the speech, but the ideal data (the same sentence in different emotions) rarely exists. *How can explicit paired supervision be provided for SPFEM when paired data is scarce?*
- **Method.** Supervision from the inputs themselves: cross-emotion consistent facial correlations as losses (ASCCL, STCCL); contrastively decoupled content and emotion representations (CDRL).
- **Results.** More accurate expression editing with better lip sync across editing architectures and benchmarks.
- **My role.** Main contributor to all three: ideas, methods, implementation, experiments and writing; my advisor

polished the writing.

Style-Consistent Scene Editing [4] 2024 – 2025

- **Motivation.** Text-guided photo edits often look out of place because their style does not match the rest of the scene. *How can an edit follow the user's intent while matching the style of the scene?*
- **Method.** A diffusion editor with parallel text and style cross-attention; style learned by contrasting regions within and across images.
- **Results.** Built a benchmark with style metrics; improved text alignment and style consistency over baselines.
- **My role.** Idea, method, benchmark, experiments and writing; my advisor polished the writing.

Publications

[1] **Breaking the Vision–Action Shortcut: Latent Interface Training for Generalizable Robot Foundation Models**
Jianman Lin, Shailesh Shailesh, Zhongyi Luo, Jiafei Duan.
arXiv:2609.12641, 2026; under review at **ICRA**. **Co-first author**

[2] **E0.5: From Reconstructive to Future-Predictive 3D Foundations for Robotic World-Action Modeling**
X-Era AI Lab Team (26 authors), with **Jianman Lin** as a core contributor.
arXiv:2610.36011, 2026.

[3] **Learning Spatial-Temporal Coherent Correlations for Speech-Preserving Facial Expression Manipulation**
Tianshui Chen, **Jianman Lin**, Zhijing Yang, Chunmei Qing, Guangrun Wang, Liang Lin.
IEEE TPAMI, 2026. **Co-first author**

[4] **Neural Scene Designer: Self-Styled Semantic Image Manipulation**
Jianman Lin, Tianshui Chen, Chunmei Qing, Zhijing Yang, Shuangping Huang, Yuheng Ren, Liang Lin.
IEEE TIP, 2025. **First author**

[5] **Contrastive Decoupled Representation Learning and Regularization for Speech-Preserving Facial Expression Manipulation**
Tianshui Chen, **Jianman Lin**, Zhijing Yang, Chunmei Qing, Yukai Shi, Liang Lin.
IJCV, 2025. **Co-first author**

[6] **Learning Adaptive Spatial Coherent Correlations for Speech-Preserving Facial Expression Manipulation**
Tianshui Chen, **Jianman Lin**, Zhijing Yang, Chunmei Qing, Liang Lin.
CVPR, 2024. *Highlight (top 2.8%)*.

Industry Experience

Research Intern: AI & Robotics Dec. 2023 – Present
X-Era AI & ExpanderaAI Lab Guangzhou, China

- **Robotic flexible packaging (2026 – Present), project lead.** Organized and designed a dual-arm system that packs items into a carton and seals it, a ∼5-minute long-horizon task with **90% success**, combining subtask reasoning with offline reinforcement learning; built the pipeline from teleoperation to deployment. [Video]
- **AI interior design (2025 – 2026).** Brought multimodal large models into a vertical business for the Feiliu AI interior design product: VLM floor-plan parsing (98% accuracy), LLM layout generation and agents.
- **Digital humans (2023 – 2024).** Model training and deployment; StyleGAN priors for few-shot, identity-consistent avatars.

Honors, Patents & Service

- **Honors:** National Scholarship (2026); CVPR 2024 Highlight; First-Class Scholarship; National First Prize, CUMCM; National Third Prize, Mechanical Innovation Design Competition.
- **Patents:** 3 granted Chinese invention patents, 2 as first inventor. **Language:** IELTS Overall **7.0**.
- **Reviewer:** ICRA, CVPR, NeurIPS, ECCV.